

Product Carbon Footprint Study Report

Product Name: cookstove, gas or electric

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: cookstove, gas or electric

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 Item cookstove

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 Item cookstove.

2. System Boundary

cookstove production, gas or electric, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of cookstove, gas or electric produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **152.550 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

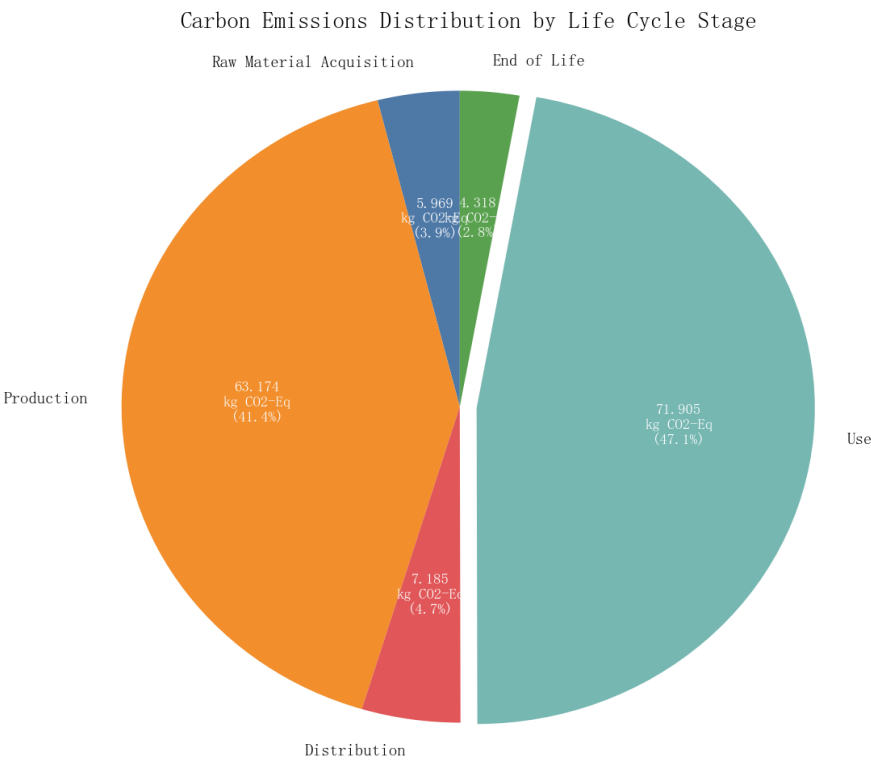


Table 1: Carbon Emissions by Life Cycle Stage

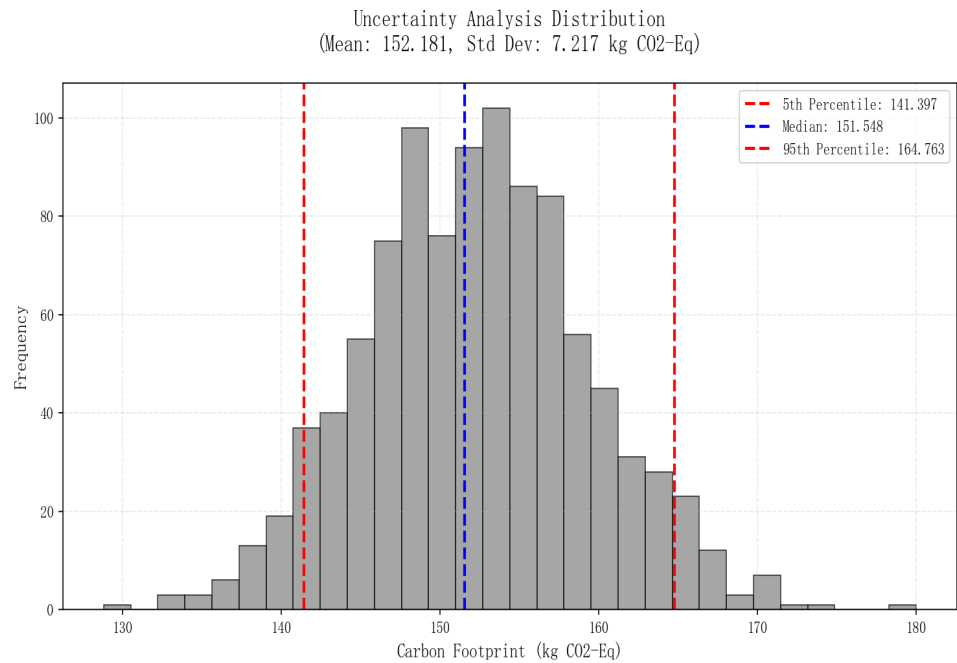
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	5.969	3.91
Production	63.174	41.41
Distribution	7.185	4.71

Use	71.905	47.14
End of Life	4.318	2.83
Total	152.550	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	152.181
Standard Deviation	7.217
Coefficient of Variation (CV)	4.74%
Median	151.548
5th Percentile	141.397
95th Percentile	164.763

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the cookstove (gas or electric) is calculated at 152.550 kg CO₂-Eq, which represents a moderate environmental impact for a household appliance over its lifecycle. The lifecycle stage contributions reveal that the "use" phase is the dominant contributor, accounting for 47.1% (71.905 kg CO₂-Eq) of total emissions, closely followed by the "production" stage at 41.4% (63.174 kg CO₂-Eq). This indicates that the majority of greenhouse gas emissions arise from energy consumption during operation and manufacturing processes, while raw material extraction (3.9%), distribution (4.7%), and end-of-life (2.8%) contribute relatively minor shares.

The high contribution from the use phase suggests that the cookstove's energy efficiency and the carbon intensity of the energy source (e.g., grid electricity or gas) are critical factors. The production stage's significant emissions likely stem from material processing, component assembly, and energy-intensive manufacturing. The low contributions from raw materials and end-of-life imply that upstream and downstream processes have less influence, though improvements in material selection and recycling could still yield benefits. Overall, the carbon footprint is concentrated in the operational and production stages, highlighting opportunities for targeted reductions.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of average grid electricity or gas composition data for the "use" phase, which may not reflect regional variations in energy sources or user behavior. The "production" stage assumes typical manufacturing processes for gas or electric cookstoves, but specific technologies (e.g., induction vs. resistive heating) or production efficiencies are not differentiated. The "distribution" phase assumes standardized transport distances and modes, while "end-of-life" assumes common disposal practices without detailed recycling rates.

Data sources and quality are limited by the reliance on generic LCA databases (e.g., Ecoinvent or similar) for emission factors, which may not capture product-specific variances. The "top process contributions" are unknown due to incomplete data, precluding identification of key sub-processes. System boundaries exclude infrastructure (e.g., power plants, gas pipelines) and user-related factors like cooking habits or maintenance, potentially underestimating indirect impacts. Additionally, the study does not account for biogenic carbon or long-term storage effects, which could influence results for gas cookstoves.

3. Suggestions for Improvement

- ****Improve use-phase efficiency****: Promote high-efficiency electric cookstoves (e.g., induction) and low-carbon energy sources (e.g., renewable electricity or biogas) to reduce the 47.1% emissions from operation. Implement energy labeling and user education on optimal cooking practices.
- ****Optimize production processes****: Adopt cleaner manufacturing technologies (e.g., electrification with renewables) and lightweight materials (e.g., recycled metals or bioplastics) to cut the 41.4% production emissions. Conduct supplier audits to enforce low-carbon sourcing.

- ****Enhance end-of-life management****: Design cookstoves for disassembly and recycling to increase material recovery rates beyond current practices, reducing the 2.8% end-of-life impact. Partner with recycling facilities for metal and electronic component recovery.
- ****Improve data quality****: Collect primary data on specific manufacturing processes, energy mix at point of use, and user behavior to replace generic assumptions. Conduct sensitivity analyses for key parameters (e.g., energy source, transport distance) to refine uncertainty bounds (CV: 4.74%).
- ****Expand system boundaries****: Include upstream infrastructure (e.g., gas extraction, power plant construction) and downstream waste treatment details to capture full life cycle impacts. Investigate top process contributions through detailed process-level data collection.